Assignment 3

QGIS and PostgreSQL/PostGIS

Shraddha Gourishetty
Faculty of Environment, University of Waterloo
GEOG 387: Spatial Database
Dr. Rob Feick
Nov 13, 2021

Table of Contents

1. ***Introduction*** *3*
2. ***Study Area*** *3*
3. ***Methods*** *4*
4. ***Data Types*** *6*
5. ***Results*** *7*
6. ***Conclusion*** *7*
7. ***References*** *9*

1. Introduction

Shapefiles remain the most widely used format for distributing geographical data. A shapefile comprises multiple files working together: the geometry data file, a file defining the geographic coordinate system, a file containing attribute characteristics, and an index file for the geometries. The primary advantage of shapefiles is their rendering speed they draw significantly faster than coverages, making them efficient for visualization and analysis.

GeoPackage represents a modern evolution in geospatial data storage. Built as a subset of SQLite, a lightweight SQL implementation designed for standalone databases, GeoPackage offers enhanced compatibility and usability for both GIS and non-GIS software. It addresses several shapefile limitations, including non-standardized coordinate reference system formats and obsolete attribute field constraints.

A database schema defines the conceptual structure of a relational database or portion thereof. It can be represented visually or through integrity constraints, which are formulas written in data definition languages like SQL that govern database behavior. Implementing multiple private schemas provides an effective method for segregating database users, particularly when handling confidential data. This approach supports user separation across distinct applications, enables multi-tenancy architectures, and facilitates application version migrations.

Highways play an essential role in making travel more convenient and comfortable for both personal and commercial purposes, as well as for transporting goods. Across the country, the highway system connects major cities and rural regions. Solid highway infrastructure is crucial for every country's successful growth and development. There was a time when even emergency responders were unreachable to individuals who lived in more remote places due to a lack of adequate roads.

When building highways, various aspects should be considered to manage impacts on communities and lands. Some of the factors that should be considered are the amount of land occupied by the highway roads, disrupted farm activities due to the segmentation of existing field plots, impacts on the environment such as forests, wetlands, and habitat, and social impacts on residences and farm buildings. This analysis examines these aspects for two alternative highway routes to determine which option minimizes environmental and community disruption.

2. Study Area

The study area consists of land in Ontario where two alternative highway routes have been proposed by the Ministry of Transportation Ontario (MTO). The analysis uses data from CanVec dataset to evaluate impacts on farmstead areas, wooded areas, and road segments.

Four critical tables form the database schema for this analysis: building_0, residential_area_2, pipeline_1, and waterbody_2. The building table identifies structures requiring demolition, providing essential information about deconstruction costs, timelines, and affected communities. This data addresses research questions such as: Which communities face displacement due to highway construction?

The residential_area table captures information about homes and residents requiring relocation. Highway construction significantly impacts residential areas through noise pollution, air pollution, and property loss. This table helps answer: What are the damaging impacts on local populations due to highway construction?

The pipeline table documents existing utility infrastructure that requires modification or complete reconstruction. Understanding pipeline systems is crucial for project planning and addresses questions like: If highway pipelines require repair or replacement, what is the construction timeline?

The waterbody table identifies water features intersecting the proposed highway corridor and informs mitigation strategies. Construction over water bodies creates long-term risks, including highway flooding during heavy rainfall events. This table supports environmental impact analysis, answering: Which highway construction damages result from environmental factors?

3. Methods

The datasets used to analyze the routes are farmstead area, highway routes (line), wooded area, highway roads (polygon), and road segments. The farmstead area dataset consists of land that comprises farm buildings in the entire study area. The highway route layer consists of data of two highway routes traced from the MTO alternate highways roads raster image. The wooded area layer consists of data of land that covers forest vegetation. The highway roads layer is the same as the highway routes layer but covers 90 meters of land around the highway routes data. The road segments layer consists of data of road intersections divided into segments.

To analyze these datasets, I used spatial queries for three tables: farmstead area, wooded area, and highway roads. For the farmstead area and wooded area, I used the ST_Crosses spatial query, and for the highway roads, I used ST_buffer.

For the farmstead area, the query used was:

```
select id, row_number () over () qgis_id, f.id as farmstead_id, f.geom
from sgourish.farmstead_area_agric_res_inventory f, sgourish.hwy_route h
where st_crosses(f.geom, h.geom)
```

This query created the layer of farmstead area that crosses the highway, shown in Figure 1 and Figure 2 in green color, mostly on the second road.

Figure 1:

Map Created Identifying Farmstead Areas and Wooded Areas

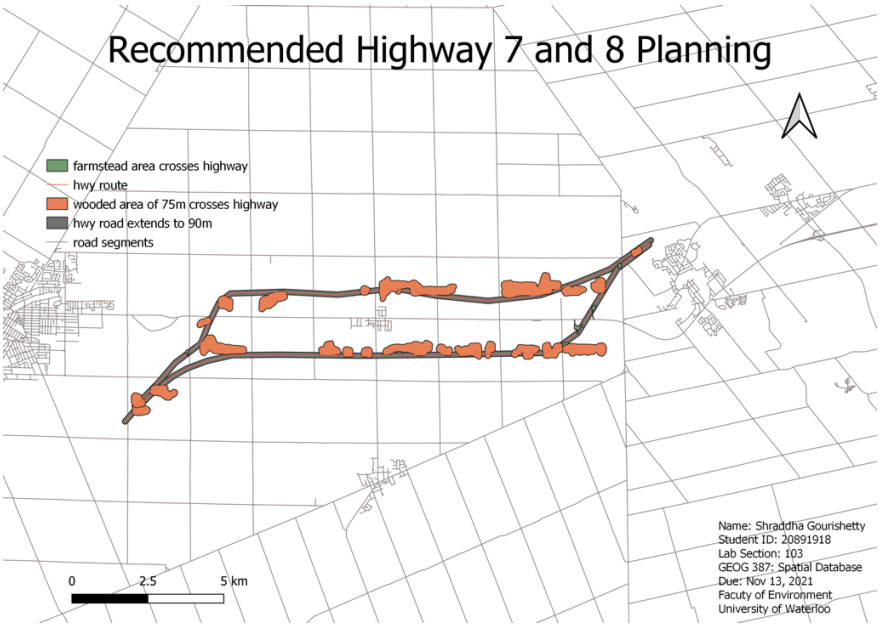

Figure 2:

A Closer View of Figure 1

Before analyzing the wooded area, I first buffered the entire wooded_area_2 table to 75 meters using the query:

```
select id, st_buffer(geom, 75)::geometry(polygon, 26917) as geom
from sgourish.wooded_area_2
```

Then I identified areas where the buffered wooded area crosses the highway routes using:

```
select row_number () over () as qgis_id, w.id as wood_id, w.geom
from sgourish.wooded_area_75m w, sgourish.hwy_route h
where st_crosses(w.geom, h.geom)
```

The resulting layer shows wooded area of 75m that crosses the highway, displayed in Figure 1 and Figure 2 in light brown color. This data gives information on how much land would need to be deforested, which will impact the environment in many ways.

For the highway routes, the query used was:

```
select id, st_buffer(geom, 90)::geometry(polygon, 26917) as geom
from sgourish.hwy_route
```

This query created the layers of highway roads extending to 90m, which covers the land of highway routes to 90 meters. This data gives information on the width of the land that will be constructed.

4. Data Types

The character varying datatype stores a variable-length string of letters, numbers, and symbols. It contains the maximum size of the column in bytes and the minimum number of bytes specified for that column. So, the "character varying (100)" can be either the maximum size of the column or the number of bytes in that column.

The double-precision data type is a numeric data type, also known as float8. In PostgreSQL, double-precision numbers are regarded as floating-point values. This means that if you try to save a number with several decimal digits, some rounding will apply. For example, if you attempted to record the outcome of 2/3, some rounding might occur once the 15th digit is reached.

Two other data types not encountered in ArcGIS are tsvector and bigserial. For full-text search, Postgres supports a data type called tsvector. A tsvector value combines alternative spellings of the same word and eliminates duplicates to produce a sorted list of unique words known as

lexemes. The bigserial data type assigns a sequential integer of the BIGINT data type when a new record is entered. The BIGSERIAL data type behaves similarly to the SERIAL data type, but with a greater range.

5. Results

Figure 1 shows the map of the two alternative highway routes along with the wooded area that covers up to 75 m of the highway routes, highway roads that cover up to 90 m, farmstead area that crosses the highway roads, and road segments.

The data collected from the given dataset analyzes what highway route is most suitable to build. From Figure 1 and Figure 2, we can see that road 2 crosses a lot more land that would need to be demolished to construct the corridor, whereas road 1 crosses less land. This is shown in Figure 2 more clearly.

The farmstead area layer that crosses the highway tells us the area that intersects the highway roads. If a lot of area is intersected, then it impacts the people who own the property in that area. The results show that road 2 intersects significantly more farmstead area than road 1.

The wooded area crossing analysis shows how much land would need to be deforested, which will impact the environment in many ways. Road 2 crosses substantially more wooded area than road 1, indicating greater environmental impact.

The highway roads layer extending to 90m shows the width of land that will be constructed. Both routes require similar widths, but the total impact differs based on what each route crosses.

6. Conclusion

In my opinion, road 1 is most suitable and should be preferred for construction. If we preferred to build road 2, many people who live in the farmstead area would need to be evacuated, which will result in loss of their property and money they invested to construct their property. Much of the wooded area, which is considered forest area, would need to have trees and vegetation cut down. This will directly impact the environment, leading to less forested area and less pure air in the atmosphere. These environmental factors can lead to extreme air pollution due to the increase in vehicle usage and affect global warming on a smaller scale in the long run, which also needs to be addressed when considering building highway roads.

More data such as the number of travelers that will use the roads, weather conditions in the area where the highway is built, and the number of lanes that will be built on these routes should also be considered as a priority, which is not given in the dataset CanVec. The number of travelers

who will use the roads is important because it will tell us whether the road is in the actual interest of the public and at what rate it will be used on a daily basis. Weather condition data will tell us how weather will impact traffic on a daily basis and the period that is needed to reconstruct the roads due to weather damages. The number of lanes that are going to be constructed in those routes will tell us the intensity of traffic depending on the time the roads will be used.

Building highway corridors has negative and positive impacts on people and the environment. To maximize positive factors, we need to look closely at various datasets. Highway construction requires comprehensive consideration of multiple interconnected factors, each carrying distinct implications for project success, community impact, and environmental sustainability. Based on the analysis conducted, route 1 will be most suitable considering multiple factors.

7. References

Häme, J. (2019, December 5). *Shapefile vs. Geojson vs. Geopackage*. Terramonitor Feed. Retrieved November 14, 2021, from https://feed.terramonitor.com/shapefile-vs-geopackage-vs-geojson/.

What is a database schema? Lucidchart. (n.d.). Retrieved November 14, 2021, from https://www.lucidchart.com/pages/database-diagram/database-schema.

Character varying(m,R) data type. IBM. (n.d.). Retrieved November 14, 2021, from https://www.ibm.com/docs/en/informix-servers/12.10?topic=types-character-varyingmr-data-type.

How to use the postgresql double precision type. ObjectRocket How to Use the PostgreSQL Double Precision Type Comments. (n.d.). Retrieved November 14, 2021, from https://kb.objectrocket.com/postgresql/how-to-use-the-postgresql-double-precision-type-1435.

Bigserial Data Type. (n.d.). Retrieved November 14, 2021, from https://help.hcltechsw.com/onedb/1.0.0.0/sqr/ids_sqr_097.html#:~:text=BIGSERIAL%20data%20typeThe%20BIGSERIAL,but%20with%20a%20larger%20range.